

Enhancing Reliability and Business Continuity in Global Fixed Income Trading Systems

Introduction

A leading global investment management firm partnered with NuWare to strengthen its Fixed Income Trading (FIT) application landscape, ensuring continuous availability, improved performance, and robust disaster recovery capabilities.

The client's trading operations spanned multiple geographies, relying on mission-critical systems for portfolio management, compliance, and trade execution. Any disruption in these systems could directly impact trading performance, data accuracy, and client satisfaction.

NuWare implemented a comprehensive Application Support and Development framework, balancing offshore efficiency with onsite responsiveness, to deliver round-the-clock reliability, scalability, and operational transparency.

The Challenge

The client's core business applications for fixed income trading and operations required a high-availability support model to handle real-time transactions across regions. However, the existing setup faced key limitations:

- 1. Lack of SLA-Driven Support:**
Mission-critical systems lacked defined service levels, resulting in inconsistent performance and potential business impact.
- 2. Batch Processing Delays:**
Inefficient batch job execution caused data bottlenecks, leading to escalations and missed deadlines.
- 3. No Disaster Recovery Framework:**
The absence of a remote disaster recovery solution made the system vulnerable in case of primary data center failures.
- 4. Data Quality Risks:**
Inconsistent data validation checks threatened decision accuracy and compliance readiness.
- 5. Limited Process Transparency:**
The existing support model offered minimal visibility into operational priorities and issue tracking.

The client needed an agile, SLA-based support and development model to guarantee system reliability, ensure data integrity, and safeguard trading continuity.

NuWare's Approach

NuWare deployed a two-tier Application Support and Development model; combining offshore scalability with onsite agility, to manage the client's Fixed Income Trading systems.

1. End-to-End Application Support Setup

- Established a core FIT team with dedicated Primary (Offshore) and Secondary (Onsite) units.
- Delivered continuous application monitoring, performance tuning, and incident management.
- Developed detailed documentation, ensuring process consistency and knowledge continuity.

2. Batch Process Optimization

- Identified and resolved recurring batch processing issues affecting critical financial applications.
- Enhanced job scheduling through automation tools such as Control-M, CRON TAB, and Autosys, ensuring timely data availability for trading operations.

3. Disaster Recovery Enablement

- Designed and implemented a remote-site disaster recovery framework to ensure operational continuity in the event of data center outages.
- Conducted simulation drills to validate failover efficiency and system readiness.

4. SLA Framework and Governance

- Established measurable SLAs for response, resolution, and recovery times.
- Introduced 99% internal service level compliance, enhancing reliability and stakeholder confidence.

5. Continuous Monitoring and Reporting

- Implemented proactive batch and system monitoring processes to detect and resolve issues before escalation.

- Enabled transparent reporting, empowering management to prioritize strategic vs. tactical improvements effectively.

Outcomes

The engagement resulted in significant improvements in operational efficiency, reliability, and data accuracy:

1. High Availability and Stability

- Achieved **99% uptime** for all mission-critical Fixed Income systems.
- Reduced incident response times through automated alerting and proactive monitoring.

2. Enhanced Disaster Recovery Readiness

- Implemented a **robust DR framework** ensuring business continuity during outages.
- Reduced risk exposure and downtime impact through simulation-tested recovery processes.

3. Improved Batch Processing Efficiency

- Automated scheduling eliminated manual delays and improved end-of-day reporting accuracy.
- Increased overall system reliability and throughput.

4. SLA-Driven Transparency

- Delivered measurable service performance aligned with business goals.
- Enabled management to effectively balance tactical enhancements with long-term IT strategy.

5. Cost-Effective Global Delivery

- Leveraged an **offshore-on-site hybrid model**, optimizing cost without compromising quality or responsiveness.

Tools and Technologies Used

- **Operating Systems:** Windows 9X/XP/NT, UNIX, Linux
- **Databases:** SQL, Sybase, Sybase IQ, Oracle, MS Access
- **Middleware:** IBM MSMQ for inter-application message transfer
- **Programming Languages:** .NET, PowerBuilder, J2EE, Shell, Perl

- **Scheduling Tools:** Control-M, CRON TAB, Autosys
- **Reporting Tools:** Crystal Reports (publishing FI data to the trading community)
- **Web Technologies:** HTML

Future Outlook

Following the success of this engagement, NuWare continues to assist the client in advancing its technology ecosystem through:

- **Automation and predictive monitoring** for early anomaly detection.
- **Migration to cloud-based infrastructure** for scalability and resilience.
- **Integration of AI-driven analytics** for real-time decision-making and performance optimization.

These initiatives are designed to future-proof the client's Fixed Income platform and enable real-time, intelligent trading operations.